Waweru Wallicestene Wakaba

Nairobi, Kenya
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Professional Summary

Software engineer working across fintech, web platforms, and data. Builds front ends, REST APIs, and the data pipelines behind AI agents using React, Next.js, Node.js, TypeScript, and Python, for teams in Kenya and clients abroad.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, SQL
Frontend: React, Next.js, TailwindCSS, Radix UI, HTML5, CSS3
Backend: Node.js, Express.js, NestJS, FastAPI, Prisma ORM
Databases: PostgreSQL, MongoDB (Mongoose), MySQL
APIs & Integrations: REST API Design, BetterAuth, JWT, Firebase
AI & Agents: Google Cloud (Vertex AI), Agent Development Kit, LLM Tool Calling, Groq LLM API
Mini Apps: M-PESA Mini Program Platform, Ant Design Mini, Alipay+ Framework
Apify Platform: Apify SDK, Crawlee, Playwright, Cheerio
Cloud & DevOps: AWS (S3), Firebase, Docker, Git, GitLab CI/CD

Experience

Software Engineer Intern – Safaricom PLC **May 2026 – Oct 2026**
API Engineering – M-PESA Mini Apps & Partner Ecosystem *Nairobi, Kenya*

- Built and maintained a **70+ component** TypeScript library, shipped as an installable package, used by Safaricom developers and external partner teams to build M-PESA Mini Apps for the M-PESA One app.
- Designed how Safaricom's customizations layer over the upstream **Ant Design Mini** library, so the fork keeps pulling upstream releases without breaking what we changed.
- Maintained the library's documentation and release pipeline so partner developers outside the organization could pick up a component and ship it without internal support.
- Built sandbox mini apps to prove out component behavior across Mini Program Studio environments before release.

Freelance Software Engineer – Upwork **Oct 2025 – Present**
Full-Stack Development & Apify Platform Actors *Remote*

- Delivered end-to-end frontend and backend solutions for international clients – Responsibly (Denmark), GO10 Limited (UK), and (Canada) – using React, Next.js, Node.js, Express, Prisma, and PostgreSQL, handling UI design, API implementation, and database modeling.
- Built and currently manage **20+ production-ready Apify actors** that power client AI systems and intelligent agents, combining API interception, DOM extraction, and third-party API integration into deployable data pipelines using Apify SDK, Crawlee, and Playwright.
- Engineered resilient infrastructure with proxy rotation, session pooling, exponential backoff retries, and batch processing – achieving **99%+ successful run rates** across large-scale data collection jobs.

ICT Intern – Kenya Revenue Authority (KRA) **June 2025 – Sept 2025**
Portal Security & Pension Module Integration *Nairobi, Kenya*

- Implemented **MFA, CAPTCHA**, and secure registration workflows during a system-wide portal upgrade, reducing unauthorized access attempts across a platform used by thousands of employees.
- Integrated pension management modules with a cross-functional team of **8+ developers**, delivering pension calculators and contribution filters through Agile sprints, code reviews, and QA testing.

IT Intern – Unity Homes (OCF Internship Program) **June 2024 – Aug 2024**
Frontend & Backend Development *Nairobi, Kenya*

- Built responsive React interfaces and Node.js REST APIs for enterprise property tools; automated internal workflows with custom scripts, reducing manual processing time and improving system reliability.

Projects

Dapi – AI Web Data Extraction Platform | *Next.js, Node.js, Apify, Crawlee, Playwright, LLM Agents* **Live** | **GitHub**

- Built an AI data extraction platform where users describe the data they need by text, file, or voice, and an **LLM agent** inspects the target site, writes the extraction code, and tests it on sample rows before running.
- Engineered a verifiable pipeline that reads a site's underlying API where possible, seals every value with a **SHA-256** hash linked to its source page, verifies row counts against the source, and self-repairs failed runs.

Soko Mbele – Market Price Forecasting for Farmers | *Next.js, TypeScript, Python, LightGBM, GitLab CI* **GitHub**

- Built a full-stack decision tool that tells Kenyan farmers where and when to sell, ranking nearby markets by net earnings after transport using government price data across **16 crops** and **75 markets**.
- Trained a **LightGBM** forecasting model on **330K+** historical price records, reaching **70%** directional accuracy on major price moves, with a weekly automated pipeline that retrains it and blocks any model that fails validation.

Earnest Motors – Car Dealership WebApp | *React*, *TypeScript*, *TanStack Query*, *Tailwind CSS*, *Django* **Live**

- Built the React frontend for a client's live car dealership site, with inventory search and filters, vehicle detail pages, and WhatsApp inquiries, integrated with a **Django** backend API.

Vesper – Social Media Management SaaS *(In Development)* | *Next.js*, *TypeScript*, *Express*, *PostgreSQL*, *Prisma*, *BetterAuth* **GitHub**

- Building a Buffer-like platform for scheduling posts across **social channels**, with a content calendar, engagement analytics, and a centralized media library behind a single dashboard.
- Implemented backend scheduling with node-cron and secure session-based auth via BetterAuth with Prisma ORM.

Pesapal RDBMS Challenge – Database Engine from Scratch | *TypeScript*, *Node.js*, *Express*, *React* **GitHub**

- Engineered a **custom SQL parser from scratch** for the Pesapal Junior Dev Challenge '26, supporting CREATE TABLE, INSERT, SELECT with WHERE, UPDATE, DELETE, and JOIN operations across 5 data types with primary keys, unique constraints, auto-increment, and indexing.
- Delivered an interactive REPL CLI, a REST API layer, and a React hotel booking demo app to showcase the engine end-to-end.

Education

KCA University **Sept 2021 – Nov 2025**

Bachelor of Science in Applied Computing (Software Development) | *Second Class Upper Division* *Nairobi, Kenya*

Certifications

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate | LFS101: Introduction to Linux | Data Structures & Algorithms | Front-end Development